

$$\boxed{a^T} \boxed{b} = \boxed{}$$

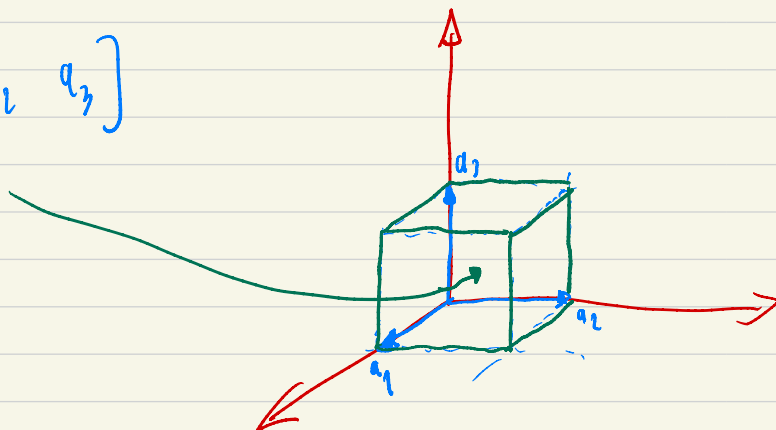
$$\boxed{a} \quad \boxed{b^T} \quad \sim \quad \boxed{}$$

$$A \Rightarrow A^T$$

$$A = \begin{pmatrix} 0 & 0 & 0 & \dots \\ 0 & 0 & 0 & \dots \\ 0 & 0 & 0 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} = A^T$$

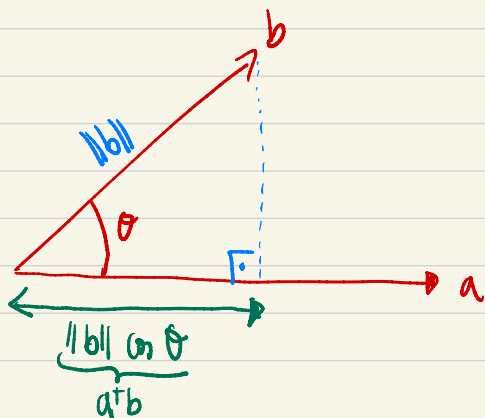
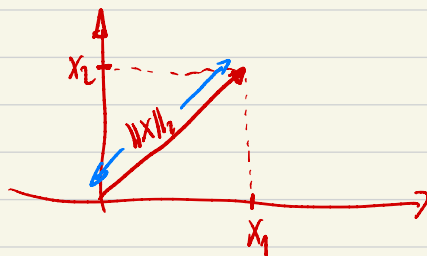
$$A = [a_1 \ a_2 \ a_3]$$

$$|\det(A)|$$



$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\|x\|_2 = \sqrt{x_1^2 + x_2^2}$$



$$a^T b = \|a\|_1 \cdot \|b\|_2 \cos \theta$$

$$\|a\|_1 = 1 \Rightarrow a^T b = \|b\|_2 \cos \theta$$

$$\|a\|_1 \neq 1 \Rightarrow \frac{a^T b}{\|a\|_1} = \|b\|_2 \cos \theta$$

